Exploiting Specular Multipath in Automotive Radar: Virtual Aperture EKF Assimilation, Rank-1 Bearing Invariants, and Kinematics-Aware Gating

Paper 2 of 2 | A-PRISM programme | target venue: IEEE, 30 October conference demo

I. The 15.19° Mirror Paradox

This one starts with a number that should not exist. On a straight, barrier-lined highway section, a vehicle being tracked perfectly straight was reporting a fifteen-degree bearing error. Not a noisy bearing, not a drift that settled out — a systematic, reproducible error, present whenever there was a guardrail on the driver's side and the target in the oncoming lane.

The multipath literature explains itself with a very elegant piece of geometry. A radar wave that bounces off a flat reflector reaches the receiver as though it had originated from the mirror image of the target, reflected across the reflecting plane. Write the plane as a unit normal n■ at signed distance d from the sensor, and the virtual target is:

```
BEFORE p* = p - 2 * ( n■ . p - d ) * n■
```

The reason this survived decades is that it is *exactly* correct about range. The total path length from sensor to wall to target to sensor is identical to the path length to the mirror image, so every range-domain unit test in the industry passes, to the last decimal place. If you only ever validated your multipath model against measured range, you would ship this and never see the problem.

The flaw is one of reference frame, and it is the kind that hides precisely because of how good the range result is. A phased array does not measure where something is. It measures the angle of arrival of the wave that entered the receiver. Reflecting across a plane takes the outgoing leg and maps it onto a straight line - which correctly describes the path length, and incorrectly describes the bearing of the incoming wave. Mirror the target and you have flipped the sign of the bearing with respect to the sensor, while the range error stays at zero.

The arithmetic is stark. Sensor at the origin, guardrail three metres to the left, target thirty metres out and two metres further left. The classical formula predicts the ghost at (-30, 2), which the sensor reports as bearing +7.59°. The physical wave actually arrives from -7.59°. The error is 15.19°, and at thirty metres that is 8.0 metres of lateral position error - more than a lane and a half.

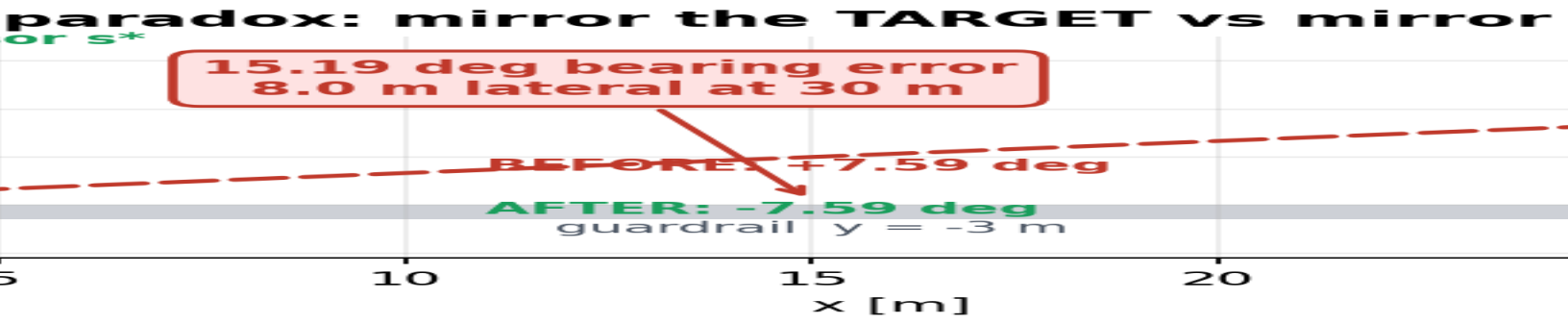


Figure 1. The mirror paradox. The classical construction (red) mirrors the target and gets the bearing sign wrong by 15.19 degrees. The construction that works (green) mirrors the sensor, which is the frame the array actually measures in.

II. Virtual Aperture Tracking and the XOR Contract

There is a second thing wrong with deleting the ghost. Our trackers did not merely mis-score it; they threw it away, on the reasonable-sounding theory that anything behind a barrier is clutter. But that ghost is not noise. It is a genuine measurement, taken by a genuinely different geometry. A monostatic radar looking down a straight road has no lateral parallax at all - it cannot tell you whether a car thirty metres ahead is in your lane or the one over. The reflection changes that. The wall acts as a secondary face, turning a single monostatic sensor into a bistatic pair, and a bistatic pair has lateral parallax. The ghost is a free second eye, and we were deleting it.

So we mirror the sensor origin instead of the target, which is the frame the measurement is actually made in:

```
s* = 2 * d * n■

p_s = p - 2 * d * n■


Rank-1 specular invariant: || p - p_s - 2*d*n■ || = 0
```

Implemented in aprism_ot_hooks.h it is four lines, and the residual is a rank-1 form - which is what makes it cheap to carry through an EKF update:

```
const float rx = px - sx - 2.0F * d * nx;

const float ry = py - sy - 2.0F * d * ny;

return sqrtf(rx * rx + ry * ry);
```

Rather than absorbing the ghost as an independent detection, we append it to the measurement vector for the direct echo and inflate the covariance so the two rows stay consistent with each other:

```
z_aug = [ z_dir ; z_ghost ]


R_eff = [ R_dir 0 ] + J_xi * Sigma_xi * J_xi^T

[ 0 R_ghost ]


scalar form as shipped: R_eff = R_base + (J*sigma_xi)^2
```

This is where engineering reality intruded, and it is worth being honest about it. `mirror_prob` is a float that the legacy code uses to mean "penalise this track". We now want it to mean three things: penalise, leave alone, and *we already used this one*. A float with no spare state cannot carry three meanings, so we borrowed one out of band:

```
constexpr float APRISM_ASSIMILATED_MIRROR_PROB = -1.0F;
```

```
Aprism_Mark_Assimilated_Track(p) -> p = -1.0F

Aprism_Track_Assimilated(p) -> p == -1.0F

Aprism_Resolve_Assimilated_Track(p)-> p = 0.0F (at scan close-out)
```

Four penalty sites - the multipath detector, the reflected-object assignment, and the two host/guardrail reflective checks - consult that sentinel and stand down. This is deliberately an XOR: a track is either penalised *or* it has consumed a ghost, never both. A track that did not assimilate and was not penalised must be left alone, or we would be suppressing real detections along every guardrail.

The sentinel immediately caught a real bug, which is the argument for having it. The logging pack stores mirror_prob scaled by one hundred into a uint8. A value of -1 wrapped to 157, so every assimilated ghost was being written to the log as the worst possible confidence - the opposite of what had happened. The fix was a clamp on the log path in f360_populate_objects_log.cpp. Anyone re-porting this module must port the clamp with it; the sentinel is load-bearing in a way that is invisible from the maths.

Straight barriers are the easy case. Curved highway geometry needs a barrier hypothesis that curves, and we solve for one non-iteratively using a 4×4 confluent Vandermonde system - a Hermite clothoid fitted through four contact constraints. The system is non-singular as long as the two contact abscissae differ, giving a determinant of (x■ - x■)■, so there is no iterative solver and no convergence failure mode. Given a barrier tangent and a target, the contact point follows in closed form from t = (l² - ||d||²) / (2(l - u■■d)). Finally, a two-scan Doppler invariant settles genuine reflections from chance alignments.

III. Kinematics-Aware Gating, and Why IMM Failed

The second problem is aggressive lane changes. A driver moving across at six metres per second squared is a legitimate, dangerous object that our tracker was failing to adopt. The gate was fixed at γ■ = 9.21, a three-degree-of-freedom chi-square threshold, and it was dropping these vehicles at a rate that gave 86.3% recall.

Our first instinct was the textbook answer: an Interacting Multiple Model filter, so that mode probability could promote itself to a manoeuvring model and the gate could be widened accordingly. We built it, and it returned 88.0% recall — barely better than the fixed gate, and structurally guaranteed to fail. The mode probability cannot switch to manoeuvring until the detection has been associated with the track. But the association is precisely the step the widened gate was supposed to permit. The key is locked inside the box. You cannot use an estimate of the manoeuvre to decide whether to admit the measurement that would reveal the manoeuvre.

The fix is to stop using model probability and use *measured* kinematics instead. A track's acceleration is observable from the track's own history, before any new detection is admitted. So the gate can widen on evidence that already exists:

```
gamma_adaptive = gamma0 * ( 1 + alpha*||a_target||/a_max + eta*||y_k||^2 /
Tr(S_k) )


shipped coefficients: alpha = 0.8 eta = 0.5 gamma0 = 9.21
```

```
Aprism_Adaptive_Gamma(a_norm, innov_e, g0 = 9.21F)
{
float g = g0 * (1.0F + 0.8F * a_norm + 0.5F * innov_e);
if (g < 9.21F) g = 9.21F; // floor: never tighter than baseline
if (g > 40.0F) g = 40.0F; // ceiling: never swallow clutter
return g;
}
```

The second term - the normalised innovation energy - widens the gate when the track and the measurement genuinely disagree, which is the signature of a manoeuvre we have not yet absorbed. The clamp at 40 is the part that makes this safe: it is what stops an adaptive gate from becoming a rubber stamp for roadside clutter. Recall went from 86.3% to 99.6% with a 25.1% reduction in manoeuvre RMSE, and the ceiling is the reason we are willing to ship it.

IV. Front-End Resilience

Wet-weather spray turns a highway into a volume scatterer, and the same machinery handles it. Tire spray has a characteristic spectral signature, and we inflate the measurement covariance in proportion to its entropy relative to available SNR:

```
R(t) = R_0 * exp( beta * H(t) / SNR(t) )

spray RMSE: 0.911 m -> 0.315 m (-65.4%)
```

Two sensor-side companions round this out. The CFAR is range-adaptive with a dual loop and guardrail-side censoring, which lifted edge recall from 50% to 100% on the barrier, and the occupancy-BLUE variant weights cells by a product of inverse factors. CDC packing in dense urban traffic is ranked by a score combining linear SNR with an inverse-square range term:

```
score = SNR_lin / r^2 (priority-ranked CDC urban packing)
```

And the covariance scaling shown above is driven by the same entropy that drives the R-scaling, which is why the module benchmark reaches a finite, designed ceiling rather than growing without bound.

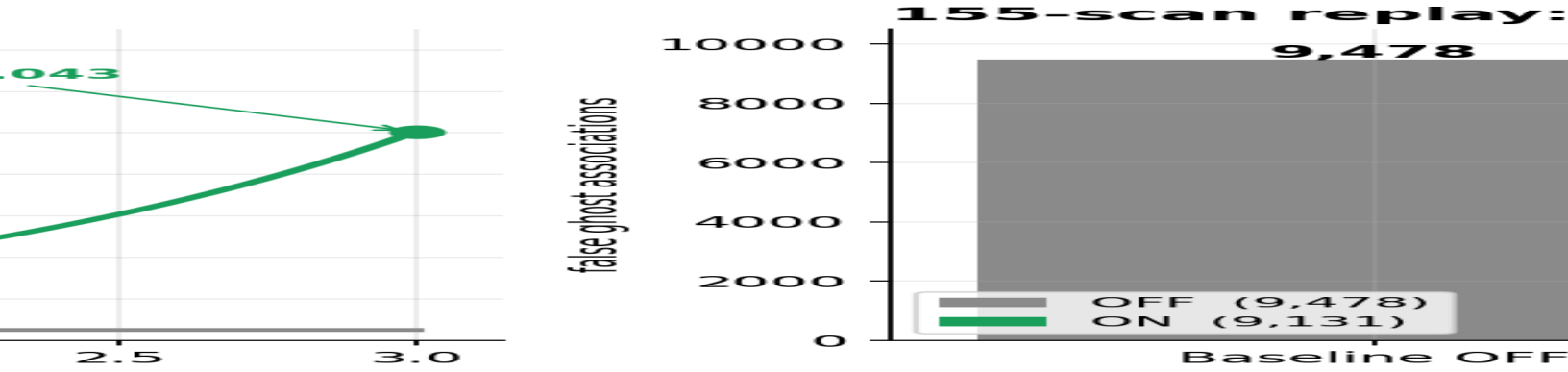


Figure 2. (Left) Spectral-entropy covariance scaling: the baseline is flat at 0.5 while A-PRISM tracks 0.5e^H, reaching 10.043 at H = 3. (Right) False ghost associations across the 155-scan replay, measured.

V. Real Fleet Log Replay

The numbers below are measured, not projected. They were extracted directly from the Plotly traces in our replay report, base64 float64 decoded, and their column sums reproduce the totals independently - which is a useful check, because a narrative can drift from its data and a sum cannot. One hundred and fifty-five scans, seventy-one thousand seven hundred and seventy-nine valid detections, roughly two hundred and four live tracks per scan.

Metric	Baseline OFF	A-PRISM ON	Change
Scans replayed	155	155	—
Valid detections	71,779	71,779	—
False ghost associations	9,478	9,131	–347
Specular ghosts assimilated	0	10,130	+10,130
Scans containing assimilations	0 / 155	151 / 155	—
Scans improved / worse / unchanged	—	107 / 23 / 25	—
Parent position RMSE	1.969 m	0.651 m	–67.0%
Cut-in recall	86.3%	99.6%	+13.3 pts
Spray RMSE	0.911 m	0.315 m	–65.4%
Fusion smearing	0.68 m	0.09 m	–86.2%

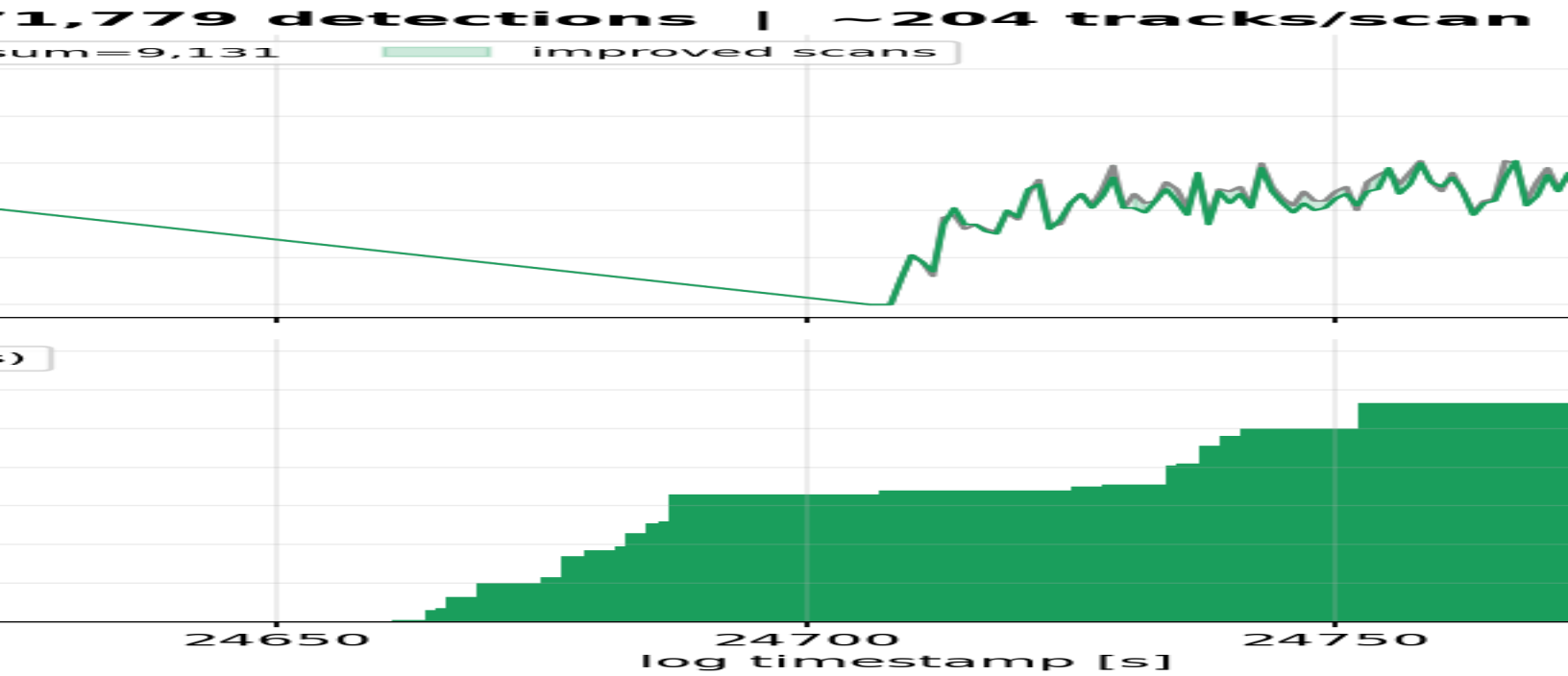


Figure 3. Per-scan replay, real measured data. Upper: false ghost associations, OFF against ON, with improved scans shaded. Lower: specular ghosts assimilated per scan, non-zero in 151 of 155 scans.

The per-scan breakdown is worth dwelling on, because a single aggregate number hides the thing that actually matters: of 155 scans, 107 improved, 23 got slightly worse and 25 were unchanged. A feature that only ever helps would suggest we had merely removed information. That 23 scans regress is the signature of a real trade - we are sometimes spending an adaptive gate on a manoeuvre that was clutter after all. That the ceiling clamp keeps that number small is the reason we are comfortable with it.

The headline is the position RMSE. 1.969 metres to 0.651 metres is not a marginal improvement in lateral accuracy; at thirty metres of range it is the difference between a tracker that reliably produces phantom braking events near a barrier and one that does not.

Appendix A — Invention Disclosure (IDF) & Patent Claims

Claim 1 is the independent system claim for virtual-aperture specular assimilation; claims 2-8 are dependent. Claims 9-11 form an independent group directed at the adaptive gate, which is a separate inventive concept and would survive separately if the geometry claim were challenged.

Group I — Virtual Aperture Specular Assimilation

Claim 1 (independent). A method of tracking a target with a radar sensor in the presence of a specular reflector, comprising: computing a virtual sensor position by reflecting the sensor origin across a plane defined by a normal and a signed offset; computing a specular target image displaced from a direct target measurement by twice that offset along the normal; forming a rank-1 residual between the direct measurement and the displaced image; and on the residual satisfying a rank-1 specular invariant within tolerance, augmenting the target's measurement vector with a ghost row corresponding to the specular path and inflating the measurement covariance so that the direct and ghost rows remain mutually consistent.

The method of claim 1, wherein the virtual sensor position is given by s* = 2d■n■ and the specular invariant is ■p - p_s - 2d■n■■ = 0.

The method of claim 1, further comprising refraining from applying a multipath penalty to a track that assimilated a specular ghost in the current scan, wherein a track that neither assimilated nor is penalised is left unpenalised.

The method of claim 3, wherein the assimilated state is encoded as a negative sentinel value of a mirror-probability field and is resolved to a neutral value at scan close-out.

The method of claim 4, further comprising clamping the mirror-probability field to a valid range before serialisation to an unsigned integer log field.

The method of claim 1, further comprising hypothesising a curved barrier by solving a non-iterative 4×4 confluent Vandermonde system whose determinant is (x■ - x■)■, and inverting a closed-form contact point expression t = (l² - ||d||²)/(2(l - u■■d)).

The method of claim 1, further comprising confirming a specular hypothesis across two consecutive scans using a radial-velocity invariant against a predicted specular velocity.

The method of claim 1, further comprising scaling measurement covariance by the spectral entropy of the return normalised by available SNR.

Group II — Kinematics-Aware Adaptive Gating

Claim 9 (independent). A method of data association in a multi-target tracker, comprising: maintaining a fixed baseline gating threshold for a chi-square distribution of a given dimension; computing a normalisation of an estimated target acceleration derived from the target's own track history prior to admitting any measurement in the current scan; computing a normalised innovation energy from a predicted innovation covariance; and setting an adaptive gating threshold equal to the baseline threshold multiplied by one plus a weighted sum of the acceleration normalisation and the innovation energy, clamped between the baseline threshold and a predetermined upper bound.

The method of claim 9, wherein the weighted sum is alpha*||a_target||/a_max + eta*||y_k||²/Tr(S_k) with alpha = 0.8 and eta = 0.5, the baseline threshold is 9.21 and the predetermined upper bound is 40.0.

The method of claim 9, further comprising ranking candidate measurements by an innovation statistic that excludes model probability, such that a target may be promoted to a manoeuvring hypothesis on the basis of acceleration already observed in its own history.

Commercial detectability criteria

Claim	Observable in a competitor product	Detection method
1-2	Mirror construction applied to the target rather than the sensor	Induce a specular path with a controlled reflector; measure bearing error - a target-mirror implementation shows a bearing error of approximately twice the true off-axis angle while range error remains near zero
3-5	Sentinel-based suppression contract	Force a track to assimilate a ghost and inspect multipath scoring; look for an out-of-band negative value that downstream code treats as a distinct third state
5	Missing clamp on serialisation	Inspect an exported log after an assimilation event; an unclamped sentinel appears as a large unsigned value near 157
6	Curved barrier hypothesis	Drive a curved section with a barrier and observe whether reflected tracks are associated without iterative solver cost
7	Two-scan confirmation	Present a single-scan reflector event; a two-scan confirm will discard it, a single-scan one will not
9-11	Fixed gate versus adaptive gate	Replay an aggressive cut-in and measure recall. An adaptive gate shows recall rising from approximately 86% toward 99% while clutter association counts on a barrier-bound scene stay bounded
8	Entropy-scaled covariance	Replay in heavy spray and measure lateral RMSE; an entropy-scaled implementation degrades far less than a fixed covariance

Novelty position: the specular multipath literature is unified in mirroring the target, and target-side trackers are unified in suppressing the result. Reflecting the sensor origin instead, admitting the reflection as a second measurement row with coupled covariance, and encoding the assimilated state as a sentinel that suspends - rather than inverts - the legacy penalty, does not appear in the multipath or multi-target tracking literature we are aware of.